

Theory of Automata

Greibach Normal Form

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Definition

- A CFG is in Greibach normal form if each rule has one of these forms:

i. $A \rightarrow aA_1A_2...A_n$

ii. $A \rightarrow a$

iii. $S \rightarrow \lambda$

where $a \in \Sigma$ and $A_i \in V - \{S\}$ for $i = 1, 2, \dots, n$

Definition

- A CFG is in Chomsky normal form if each rule has one of these forms:

i. $A \rightarrow BC$

ii. $A \rightarrow a$

iii. $S \rightarrow \lambda$

where $B, C \in V - \{S\}$

Conversion

- Convert from Chomsky to Greibach in two steps:
 1. From Chomsky to intermediate grammar
 - a. Eliminate direct left recursion
 - b. Use $A \rightarrow uBv$ rules transformations to improve references (explained later)
 2. From intermediate grammar into Greibach

Eliminate direct left recursion

- Before

$$A \rightarrow A\underline{a} \mid \mathbf{b}$$

- After

$$A \rightarrow \mathbf{b}Z \mid \mathbf{b}$$

$$Z \rightarrow \underline{a}Z \mid \underline{a}$$

- Remove the rule with direct left recursion, and create a new one with recursion on the right

Eliminate direct left recursion

- Before

$A \rightarrow A\underline{a} \mid A\underline{b} \mid \mathbf{b} \mid \mathbf{c}$

- After

$A \rightarrow \mathbf{b}\underline{Z} \mid \mathbf{c}\underline{Z} \mid \mathbf{b} \mid \mathbf{c}$

$Z \rightarrow \underline{a}Z \mid \underline{b}Z \mid \underline{a} \mid \underline{b}$

- Remove the rules with direct left recursion, and create new ones with recursion on the right

Eliminate direct left recursion

- Before

$$A \rightarrow A\underline{B} \mid \mathbf{BA} \mid \mathbf{a}$$

$$B \rightarrow b \mid c$$

- After

$$A \rightarrow \mathbf{BA}\underline{Z} \mid \mathbf{a}\underline{Z} \mid \mathbf{BA} \mid \mathbf{a}$$

$$Z \rightarrow \underline{BZ} \mid \underline{B}$$

$$B \rightarrow b \mid c$$

Transform $A \rightarrow uBv$ rules

- Before

$$A \rightarrow uBb$$

$$B \rightarrow w_1 / w_2 / \dots / w_n$$

- After

$$\text{Add } A \rightarrow uw_1b / uw_2b / \dots / uw_nb$$

$$\text{Delete } A \rightarrow uBb$$

Conversion: Step 1

- Goal: construct intermediate grammar in this format

i. $A \rightarrow aw$

ii. $A \rightarrow Bw$

iii. $S \rightarrow \lambda$

where $w \in V^*$ and B comes after A

Conversion: Step 1

- Assign a number to all variables starting with S, which gets 1
- Transform each rule following the order according to given number from lowest to highest
 - Eliminate direct left recursion
 - If RHS of rule starts with variable with lower order, apply $A \rightarrow uBb$ transformation to fix it

Conversion: Step 2

- Goal: construct Greibach grammar out of intermediate grammar from step 1
- Fix $A \rightarrow Bw$ rules into $A \rightarrow aw$ format
 - After step 1, last original variable should have all its rules starting with a terminal
 - Working from bottom to top, fix all original variables using $A \rightarrow uBb$ transformation technique, so all rules become $A \rightarrow aw$
- Fix introduced recursive rules same way

Conversion Example

- Convert the following grammar from Chomsky normal form, into Greibach normal form

1. $S \rightarrow AB \mid \lambda$

2. $A \rightarrow AB \mid CB \mid a$

3. $B \rightarrow AB \mid b$

4. $C \rightarrow AC \mid c$

Conversion Strategy

- Goal: transform all rules which RHS does not start with a terminal
- Apply two steps conversion
- Work rules in sequence, eliminating direct left recursion, and enforcing variable reference to higher given number
- Fix all original rules, then new ones

Step 1: S rules

- Starting with S since it has a value to of 1
- $S \rightarrow AB \mid \lambda$
- S rules comply with two required conditions
 - There is no direct left recursion
 - Referenced rules A and B have a given number higher than 1. A corresponds to 2 and B to 3.

Step 1: A rules

- $A \rightarrow A\underline{B} \mid \mathbf{CB} \mid \mathbf{a}$
- Direct left recursive rule $A \rightarrow AB$ needs to be fixed.
Other A rules are fine
- Apply direct left recursion transformation
$$A \rightarrow \mathbf{CB}\underline{\underline{R_1}} \mid \mathbf{a}\underline{\underline{R_1}} \mid \mathbf{CB} \mid \mathbf{a}$$
$$R_1 \rightarrow \underline{B}R_1 \mid \underline{B}$$

Step 1: B rules

- $B \rightarrow \underline{A}B \mid b$
- $B \rightarrow AB$ rule needs to be fixed since B corresponds to 3 and A to 2. B rules can only have on their RHS variables with number equal or higher. Use $A \rightarrow uBb$ transformation technique
- $B \rightarrow \underline{CB}R_1B \mid \underline{a}R_1B \mid \underline{CB}B \mid \underline{a}B \mid b$

Step 1: C rules

- $C \rightarrow \underline{A}C \mid c$
- $C \rightarrow AC$ rule needs to be fixed since C corresponds to 4 and A to 2. Use same $A \rightarrow uBb$ transformation technique
- $C \rightarrow \underline{CB}R_1C \mid \underline{a}R_1C \mid \underline{CB}C \mid \underline{a}C \mid c$
- Now variable references are fine according to given number, but we introduced direct left recursion in two rules...

Step 1: C rules

- $C \rightarrow C\underline{B}R_1\underline{C} \mid \mathbf{a}R_1\mathbf{C} \mid C\underline{B}C \mid \mathbf{a}C \mid c$
- Eliminate direct left recursion
$$C \rightarrow \mathbf{a}R_1\mathbf{C}\underline{R}_2 \mid \mathbf{a}C\underline{R}_2 \mid c\underline{R}_2 \mid \mathbf{a}R_1\mathbf{C} \mid \mathbf{a}C \mid c$$
$$R_2 \rightarrow \underline{B}R_1\underline{C}R_2 \mid \underline{B}C\underline{R}_2 \mid \underline{B}R_1\underline{C} \mid \underline{B}C$$

Step 1: Intermediate grammar

- $S \rightarrow AB \mid \lambda$
- $A \rightarrow CBR_1 \mid aR_1 \mid CB \mid a$
- $B \rightarrow CBR_1B \mid aR_1B \mid CBB \mid aB \mid b$
- $C \rightarrow aR_1CR_2 \mid aCR_2 \mid cR_2 \mid aR_1C \mid aC \mid c$
- $R_1 \rightarrow BR_1 \mid B$
- $R_2 \rightarrow BR_1CR_2 \mid BCR_2 \mid BR_1C \mid BC$

Step 2: Fix starting symbol

- Rules S, A, B and C don't have direct left recursion, and RHS variables are of higher number
- All C rules start with terminal symbol
- Proceed to fix rules B, A and S in bottom-up order, so they start with terminal symbol.
- Use $A \rightarrow uBb$ transformation technique

Step 2: Fixing B rules

- Before

$$B \rightarrow \underline{C}BR_1B \mid aR_1B \mid \underline{C}BB \mid aB \mid b$$

- After

$$B \rightarrow aR_1B \mid aB \mid b$$

$$B \rightarrow \underline{aR_1}\underline{CR_2}BR_1B \mid \underline{aCR_2}BR_1B \mid \underline{cR_2}BR_1B \mid \underline{aR_1}\underline{C}BR_1B \mid \underline{aC}BR_1B \mid \underline{c}BR_1B$$

$$B \rightarrow \underline{aR_1}\underline{CR_2}BB \mid \underline{aCR_2}BB \mid \underline{cR_2}BB \mid \underline{aR_1}\underline{C}BB \mid \underline{aC}BB \mid \underline{c}BB$$

Step 2: Fixing A rules

- Before

$$A \rightarrow \underline{C}\mathbf{B}R_1 \mid aR_1 \mid \underline{C}\mathbf{B} \mid a$$

- After

$$A \rightarrow aR_1 \mid a$$

$$A \rightarrow \underline{aR_1}\underline{C}R_2\mathbf{B}R_1 \mid \underline{aC}R_2\mathbf{B}R_1 \mid \underline{c}R_2\mathbf{B}R_1 \mid \underline{aR_1}\underline{C}\mathbf{B}R_1 \mid \underline{aC}\mathbf{B}R_1 \mid \underline{c}\mathbf{B}R_1$$

$$A \rightarrow \underline{aR_1}\underline{C}R_2\mathbf{B} \mid \underline{aC}R_2\mathbf{B} \mid \underline{c}R_2\mathbf{B} \mid \underline{aR_1}\underline{C}\mathbf{B} \mid \underline{aC}\mathbf{B} \mid \underline{c}\mathbf{B}$$

Step 2: Fixing S rules

- Before

$$S \rightarrow \underline{A}B \mid \lambda$$

- After

$$S \rightarrow \lambda$$

$$S \rightarrow \underline{a}R_1B \mid \underline{a}B$$

$$S \rightarrow \underline{a}R_1\underline{C}R_2\underline{B}R_1B \mid \underline{a}C\underline{R}_2\underline{B}R_1B \mid \underline{c}R_2\underline{B}R_1B \mid \underline{a}R_1\underline{C}B\underline{R}_1B \mid \underline{a}C\underline{B}R_1B \mid \underline{c}B\underline{R}_1B$$

$$S \rightarrow \underline{a}R_1\underline{C}R_2\underline{B}B \mid \underline{a}C\underline{R}_2\underline{B}B \mid \underline{c}R_2\underline{B}B \mid \underline{a}R_1\underline{C}B\underline{B} \mid \underline{a}C\underline{B}B \mid \underline{c}B\underline{B}$$

Step 2: Complete conversion

- All original rules S, A, B and C are fully converted now
- New recursive rules need to be converted next
$$R_1 \rightarrow BR_1 \mid B$$
$$R_2 \rightarrow BR_1CR_2 \mid BCR_2 \mid BR_1C \mid BC$$
- Use same $A \rightarrow uBb$ transformation technique replacing starting variable B

Conclusions

- After conversion, since B has 15 rules, and R_1 references B twice, R_1 ends with 30 rules
- Similar for R_2 which references B four times. Therefore, R_2 ends with 60 rules
- All rules start with a terminal symbol (with the exception of $S \rightarrow \lambda$)
- Parsing algorithms top-down or bottom-up would complete on a grammar converted to Greibach normal form

Comparison of Normal forms

$$S \rightarrow Sa\mathbf{B} \mid aB$$

$$B \rightarrow bB \mid \lambda$$

- Adding a non-recursive start symbol S' and removing λ and chain rules yields

$$S' \rightarrow Sa\mathbf{B} \mid Sa \mid aB \mid a$$

$$S \rightarrow Sa\mathbf{B} \mid Sa \mid aB \mid a$$

$$B \rightarrow bB \mid b$$

- Chomsky Normal form is obtained as:

$$S' \rightarrow ST \mid SA \mid \mathbf{AB} \mid a$$

$$S \rightarrow ST \mid SA \mid \mathbf{AB} \mid a$$

$$B \rightarrow CB \mid b$$

$$T \rightarrow CB$$

$$A \rightarrow a$$

$$C \rightarrow b$$

- Greibach Normal form is obtained as:

$$S' \rightarrow aBZT \mid aZT \mid aBT \\ \mid aT \mid aBZA \mid aZA \mid aBA \\ \mid aA \mid aB \mid a$$

$$S \rightarrow aBZ \mid aZ \mid aB \mid a$$

$$B \rightarrow bB \mid b$$

$$T \rightarrow bB$$

$$A \rightarrow a$$

$$C \rightarrow b$$

$$Z \rightarrow aBZ \mid aZ \mid aB \mid a$$

abaaba

G

CNF

GNF

$S \Rightarrow SaB$
 $\Rightarrow SaBaB$
 $\Rightarrow SaBaBaB$
 $\Rightarrow aBaBaBaB$
 $\Rightarrow abaBaBaBaB$
 $\Rightarrow abaBaBaB$
 $\Rightarrow abaaBaB$
 $\Rightarrow abaabBaB$
 $\Rightarrow abaabaB$
 $\Rightarrow abaaba$

$S \Rightarrow SA$
 $\Rightarrow STA$
 $\Rightarrow SATA$
 $\Rightarrow ABATA$
 $\Rightarrow aBATA$
 $\Rightarrow abATA$
 $\Rightarrow abaTA$
 $\Rightarrow abaABA$
 $\Rightarrow abaaBA$
 $\Rightarrow abaabA$
 $\Rightarrow abaaba$

$S \Rightarrow aBZA$
 $\Rightarrow abZA$
 $\Rightarrow abaZA$
 $\Rightarrow abaaBA$
 $\Rightarrow abaabA$
 $\Rightarrow abaaba$

Examples

- Convert to GNF
- $S \rightarrow aSX \mid b$
- $X \rightarrow Xb \mid a$